[10] Why Accelerated Descent Escapes Saddles Faster

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Accelerated Gradient Descent Escapes Saddle Points Faster than Gradient Descent — RQ2, RQ3

Problem: Perturbed GD can find second-order stationary points in roughly $\widetilde{O}(\epsilon^{-2})$ gradient steps; can a practical momentum-style method provably do better? **Method:** Modify Nesterov acceleration with perturbations and a mechanism that handles highly nonconvex regions. **Main result:** An approximate second-order stationary point in $\widetilde{O}(\epsilon^{-7/4})$ gradient iterations under global gradient/Hessian smoothness assumptions.

1. Begin with the obstacle: first-order versus second-order stationarity

For differentiable $f : \mathbb{R}^d \to \mathbb{R}$, ordinary GD is

$$x_{t+1} = x_t - \eta \nabla f(x_t).$$

If $\|\nabla f(x)\|$ is tiny, the update is tiny. But $f(u, v) = u^2 - v^2$ has $\nabla f(0, 0) = 0$ and $\lambda_{\min}(\nabla^2 f(0, 0)) = -2$. The origin is a *strict saddle*. We therefore want both

$$\|\nabla f(x)\| \le \epsilon, \qquad \lambda_{\min}(\nabla^2 f(x)) \ge -\sqrt{\rho\epsilon},$$

where ρ bounds the rate of change of the Hessian. This is the standard scaled approximate SOSP target (constants suppressed).

2. First principles: why momentum can help

A simplified accelerated step maintains a displacement v_t:

$$y_t = x_t + (1 - \alpha)v_t, \qquad x_{t+1} = y_t - \eta \nabla f(y_t), \qquad v_{t+1} = x_{t+1} - x_t.$$

The paper's actual algorithm includes safeguards and perturbation logic; these equations illustrate the accelerated-dynamics core, not its full pseudocode. Contrast a GD step, which only uses the current gradient, with momentum, which retains directional movement from past updates.

3. Why a Hamiltonian is the key analytic idea

An ordinary nonconvex objective $f(x_t)$ need not decrease at every accelerated step, even if progress is being made. Introduce a Lyapunov/Hamiltonian-like energy

$$E_t = f(x_t) + \frac{\|v_t\|^2}{2\eta}.$$

The kinetic term represents stored momentum. The proof controls a suitable energy along the modified algorithm even when the function value alone is nonmonotonic. One can then show that either enough energy decreases, or the iterates are localized and local geometric information becomes useful—the paper's *improve-or-localize* principle.

4. What happens near a saddle?

Use the local Taylor expansion around $\tilde{x}$:

$$f(\tilde{x} + z) \approx f(\tilde{x}) + \nabla f(\tilde{x})^\top z + \tfrac{1}{2} z^\top H z, \quad H = \nabla^2 f(\tilde{x}).$$

Along a Hessian eigenvector u with negative eigenvalue $-\gamma$, the local gradient points toward *increasing* displacement in magnitude. Plain GD multiplies that component by approximately $(1 + \eta\gamma)$; accelerated dynamics can grow the unstable component more rapidly. Controlled perturbations keep the algorithm from remaining almost exactly on the stable manifold.

Main result and comparison. Under Lipschitz gradient, Lipschitz Hessian, lower bounded objective, and suitable algorithm parameters, the authors prove an ϵ-SOSP in roughly

$$\boxed{\text{perturbed accelerated GD: } \widetilde{O}(\epsilon^{-7/4}) \quad \text{vs. perturbed GD: } \widetilde{O}(\epsilon^{-2})}.$$

The $\widetilde{O}$ hides logarithms and problem-parameter dependencies; this is an oracle/iteration bound, not a promise about GPU wall-clock time.

5. Assumptions, limitations and interpretation

The theory is for smooth nonconvex functions with globally controlled derivatives, not arbitrary nonsmooth ReLU training. An SOSP need not be global or even a true local minimum when the landscape has degenerate saddles. The guarantee applies to the *modified* accelerated method, not automatically to every implementation of Nesterov momentum. **Our takeaway.** Momentum can have a rigorous saddle-escape advantage rather than being just a practical heuristic. But getting to an SOSP faster is not a global-optimality or generalization guarantee. **Source:** https://proceedings.mlr.press/v75/jin18a.html.